

To: Assoc. Prof. Jennifer Horkoff

Email: jenho@chalmers.se

Subject: Application for Doctoral Student in Intentional Monitoring of Agentic AI Software — Ref 248516 — Nauval Zulfikar

Dear Assoc. Prof. Horkoff,

I am writing to express my interest in the Doctoral Student position in Intentional Monitoring of Agentic AI Software at Chalmers University of Technology, as advertised. I am eager to contribute to your research on agentic AI systems and conceptual modelling, and I am available to start on 1 September 2026.

Your paper, *[TODO: verify supervisor paper title via Google Scholar]*, particularly resonates with my research interests. The exploration of goal models and intentional modelling within AI systems aligns closely with my previous work and inspires my application.

I have a strong background in data science and AI, with a focus on agentic systems and goal-driven architectures. My recent project, *Mini Digital Twin (Integrated Passenger-Freight Transport)*, involved developing a three-layer digital twin prototype with live dashboards and interactive features, demonstrating my capability in system architecture and simulation. Additionally, my project, *LLM-Generated Adaptive Shipper Decision Rules (LLM-Shipper-Profiles)*, showcases my experience with LLM applications and reproducibility, which are relevant to the themes of your research.

While my MSc is in Business Analytics, not Computer Science, I have honed my technical skills through various projects and roles, such as my work at the Public Works & Spatial Planning Department, where I led data-driven initiatives and automated ETL pipelines. This experience has equipped me with the necessary skills in software engineering for AI systems, despite the academic gap.

I have attached my CV, research statement, and transcripts for your review. I am enthusiastic about the opportunity to contribute to your research group and am confident that my skills and experiences align well with the requirements of this PhD position.

Thank you for considering my application. I look forward to the possibility of discussing my application further.

Sincerely,

Nauval Zulfikar

zulfikar.nauval1998@gmail.com | +44 7300 469048 | nauvalzulfikar.vercel.app